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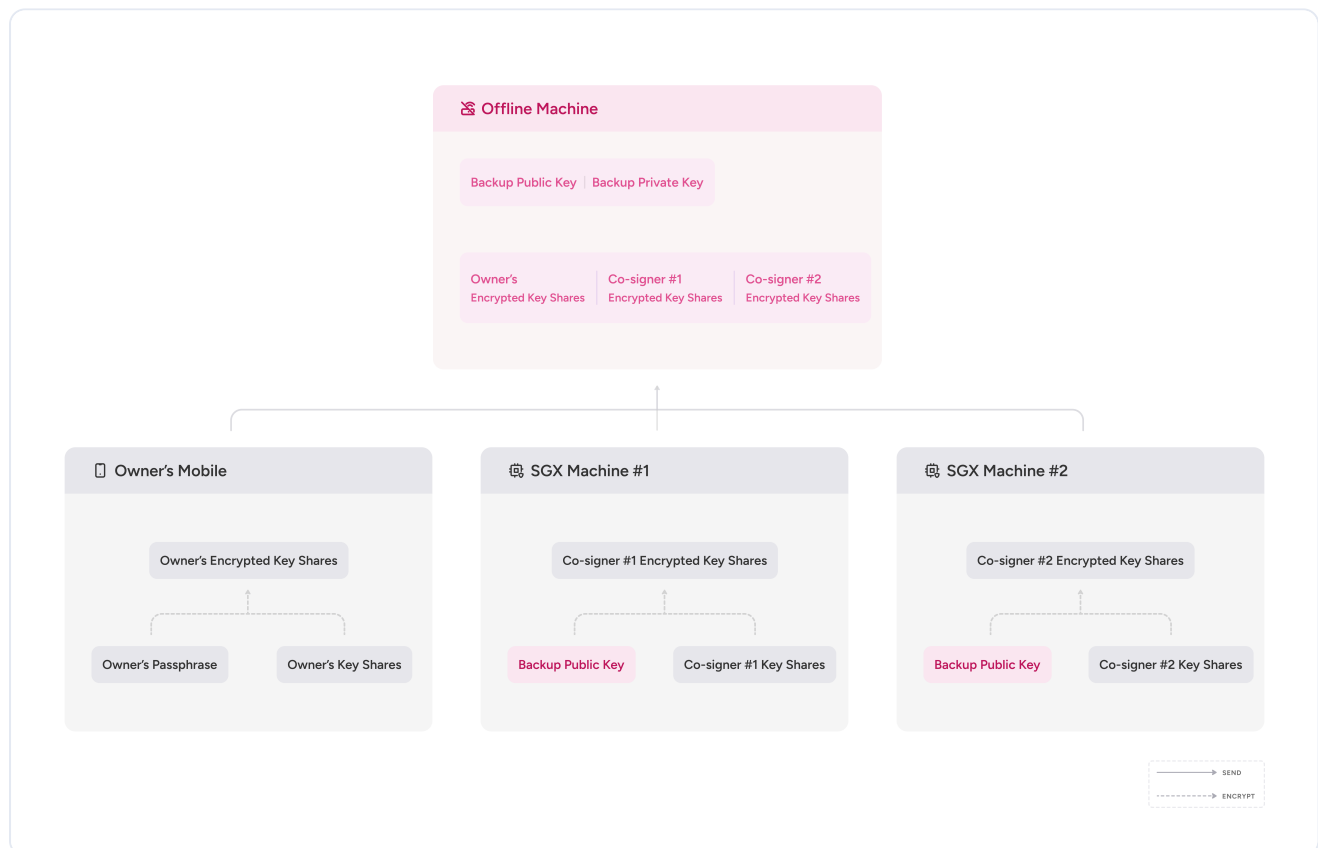
[Backup & Recovery](#)

Native workspace key backup solution

Generating a Workspace Key Backup Package - Fireblocks Recovery Utility

Overview

The diagram below illustrates how you will receive the Workspace Key Backup Package at the end of the process.



The package is composed of six files that contain the following key share components:

- ECDSA cloud key share 1 (encrypted with the RSA public key you provide)

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Important

We do not recommend running tests using your production workspace keys as it could put your signing keys at risk of exposure. You should always test the native Workspace Key Backup and Recovery process in a testnet workspace first. Contact your [Customer Success Manager](#) if you have not already obtained one.

Start the backup process

To generate the key backup package, follow the steps below.

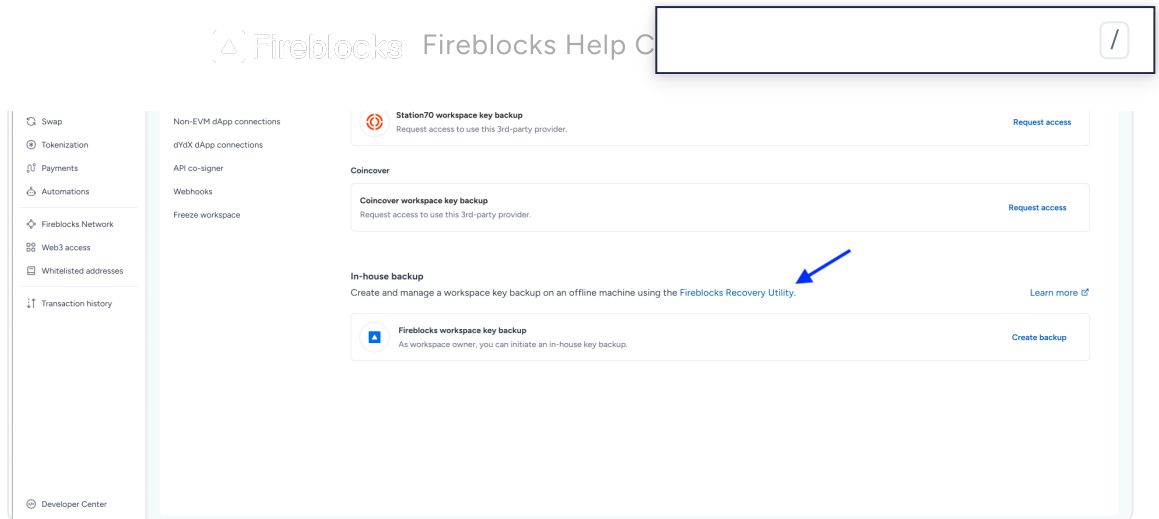
Note that throughout the key backup process, you will be alternating between multiple devices both offline and online, including using the Recovery Utility app on an air-gapped machine and the Fireblocks mobile app and Console.

To review the actual workflow, see the following video.

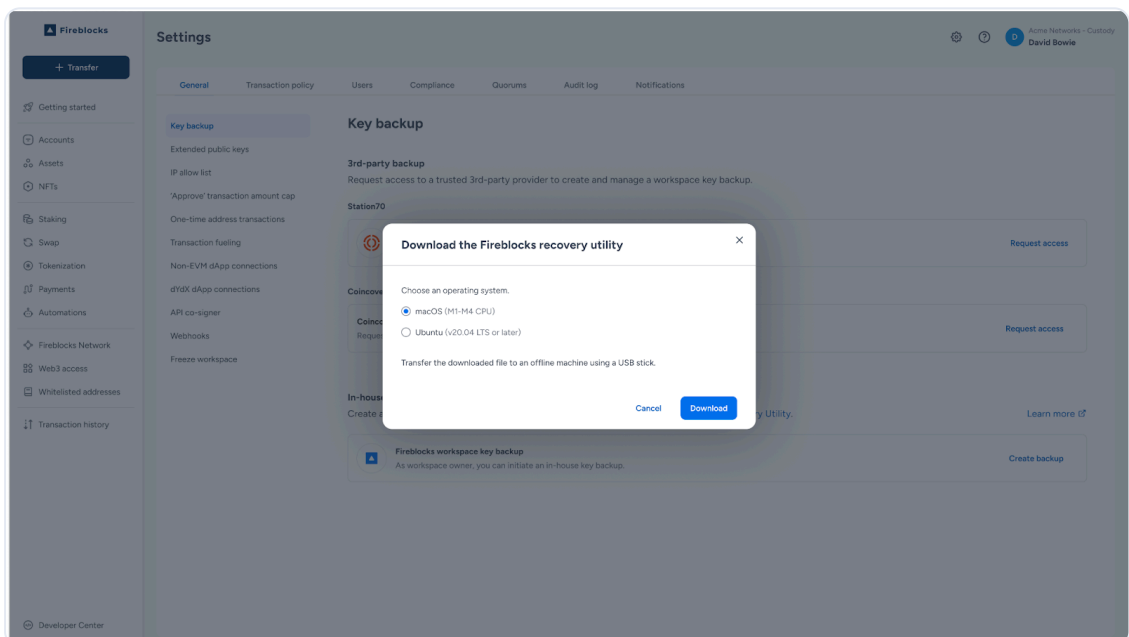


1. To start the key backup process:
 - a. Download the Fireblocks Recovery Utility application by selecting the link in **Settings > Key backup > In-house backup** section:

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and select your matching operating system:



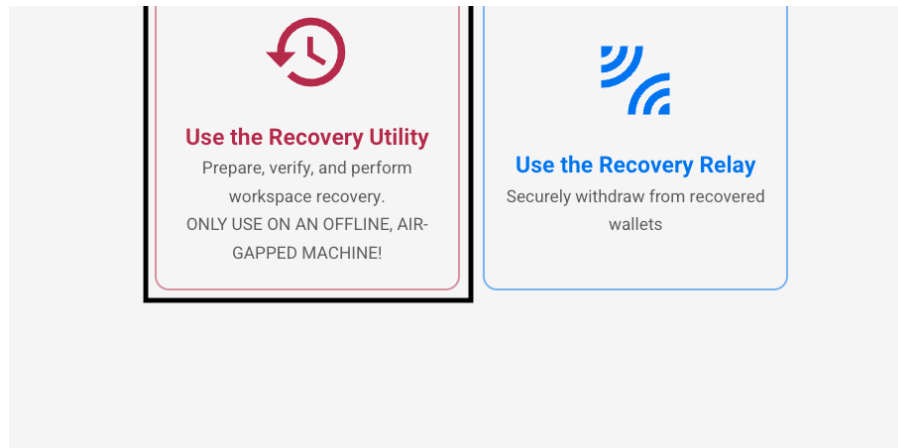
- Set up the offline recovery machine. Make sure it is air-gapped and offline.
- Transfer the Recovery Utility app (i.e., via a USB stick) to the offline machine so you can run it.

- To generate your key backup package, select **Use the Recovery Utility > Generate Keys**, and then follow the on-screen instructions.

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Warning

If your machine is not offline, a red warning message appears at the top of the Recovery Utility app window, indicating you must go offline. Keep this machine permanently disconnected from all networks.

- i. **Using the recovery utility:** enter a private key passphrase, consisting of at least four characters, and select **Generate Recovery Keys**.

Important

Make sure you save this passphrase as you will later need it to construct the full package.

For enhanced security, it is recommended to use a passphrase at least 12 characters long and include uppercase and lowercase letters, numbers, and special symbols.

2

Generate the recovery keypair

The recovery keypair is used as part of the key backup process to encrypt (and decrypt during disaster recovery) the key shares that form your private key.

Store your recovery private key and its passphrase redundantly in a secure location! You will need them to recover the assets you have stored in your Fireblocks wallets.

Recovery Private Key Passphrase



Generate Recovery Keys

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backup during a recovery. We recommend you memorize the key pair passphrase but also keep a single copy of it in a separate, secure place like a physical safe.



```
openssl genrsa -aes128 -out fb-recovery-  
prv.pem 4096
```

- ii. Extract the recovery public key (**fb-recovery-pub.pem**) from fb-recovery-prv.pem with the following command:



```
openssl rsa -in fb-recovery-prv.pem -  
outform PEM -pubout -out fb-recovery-  
pub.pem
```

- iii. Drop the public key into the recovery utility.

OR

Supply a public key

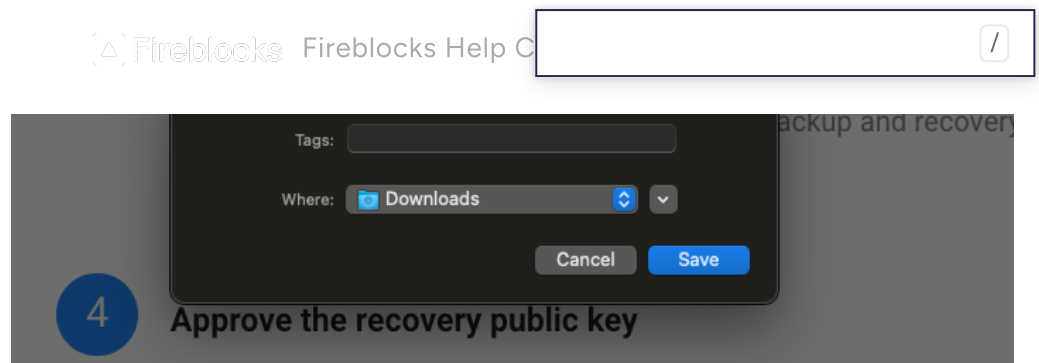
If you already have a private-public key pair, you can simply upload the public key here instead of generating a public key



Drop or select file

- iii. Select **Download Keys Zip** to download your Private Key (which must be saved on your air-gapped machine) and your Public Key

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- iv. Extract the Public Key from the zip file to an online machine with access to your Console.
- v. In your Console, connect to the specific workspace for which you want to generate the key backup package. In the General tab, under Settings, select **Create backup**.

Extended public keys

The public keys and private keys of all of wallets in this workspace are derived from the extended public keys and private keys. [Learn more](#)

Key backup

In-house key backup

You haven't backed up your keys. A backup is crucial to maintain full asset custodianship. [Learn more](#)

Create a key backup

The workspace owner must create a key backup.

Create backup

- vi. Complete the prerequisite steps to prepare the recovery Public Key. Check the **I'm ready to upload the public key file** box.
- vii. Select the Public Key file that you generated above using the Recovery Utility app, then select **Upload key**.

Create a key backup



Start the key backup process by uploading a public key file to encrypt your keys.

[Learn more](#)

Public key file (PEM)

 Drop your files here, or [browse](#)



We recommend creating a key backup only after there is an admin quorum in place [Learn more](#)

Cancel

Upload key 

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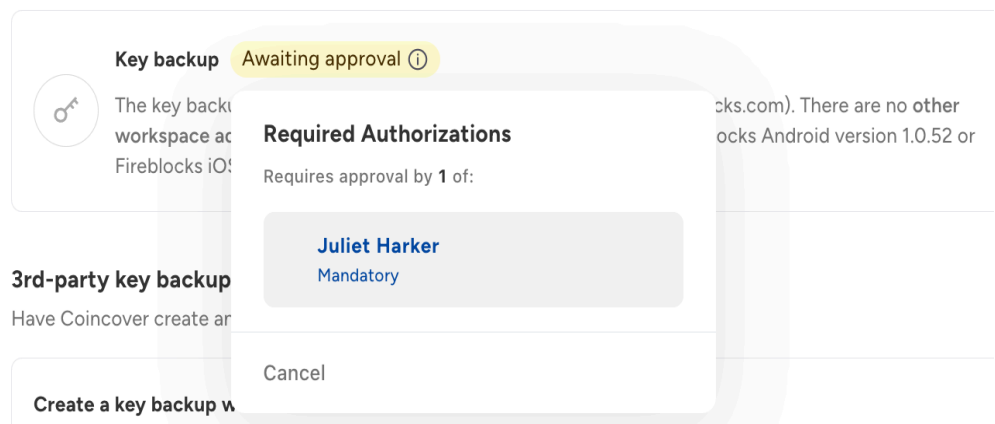
mobile app. If they do not approve within 48 hours, you must restart this process.

- x. While the request is pending approval, selecting the yellow **Awaiting approval** badge shows the approval requests and which Admins can still approve them.

Key backup

In-house key backup

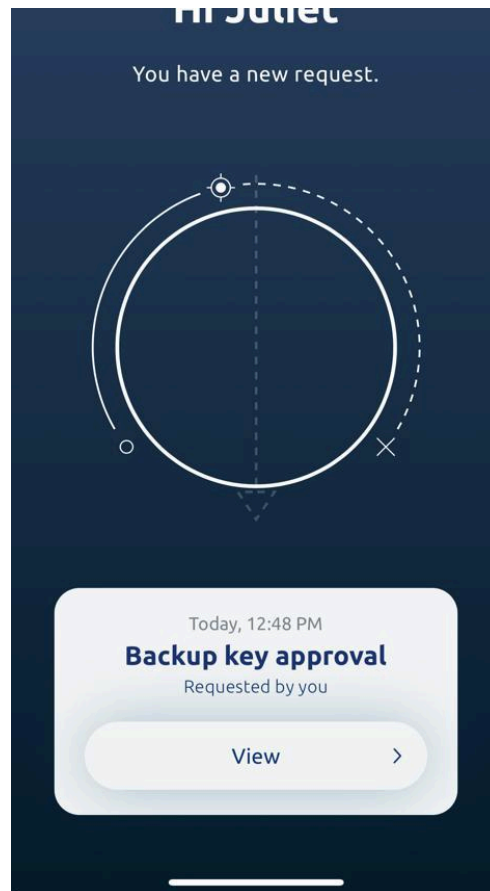
You have a key backup in progress.



- xi. Return to the air-gapped machine to follow the Recovery Utility app instructions and select **Start Approval**. This prompts you to either scan a QR code or enter a short key into your Fireblocks mobile app.
- xii. Your Admin Quorum collaborates with the workspace Owner to verify and approve the key backup by following the prompts on the Fireblocks mobile app:

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- ii. The Admin Quorum selects one of the approval methods - scan a QR code or input a short key - and informs the Owner. Since the Owner has access to the offline machine, they can proceed with the QR code option in the Recovery Utility app, while the Admins can ask the Owner for a short key.
- iii. Scan the QR code or input the short key according to the suitable method.
- iv. Once the verification is successful, the Fireblocks mobile app prompts that the key was verified.

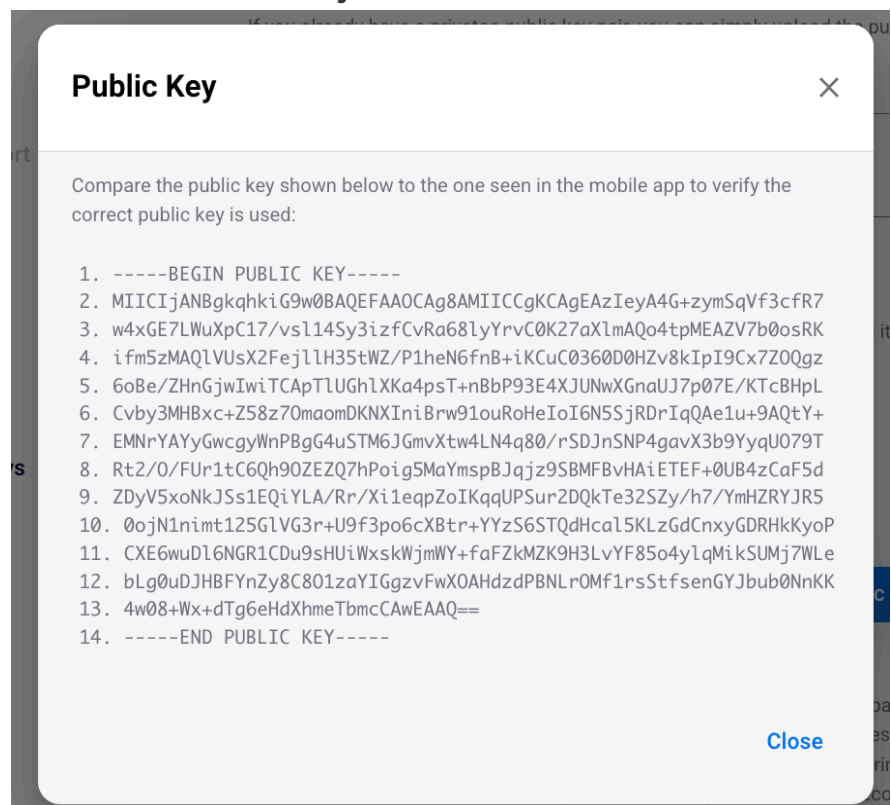
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- vi. To view the public key on the offline machine, go back to the Recovery Utility app on the air-gapped machine and select **View Public Key**.



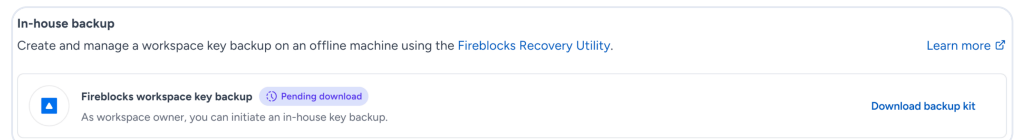
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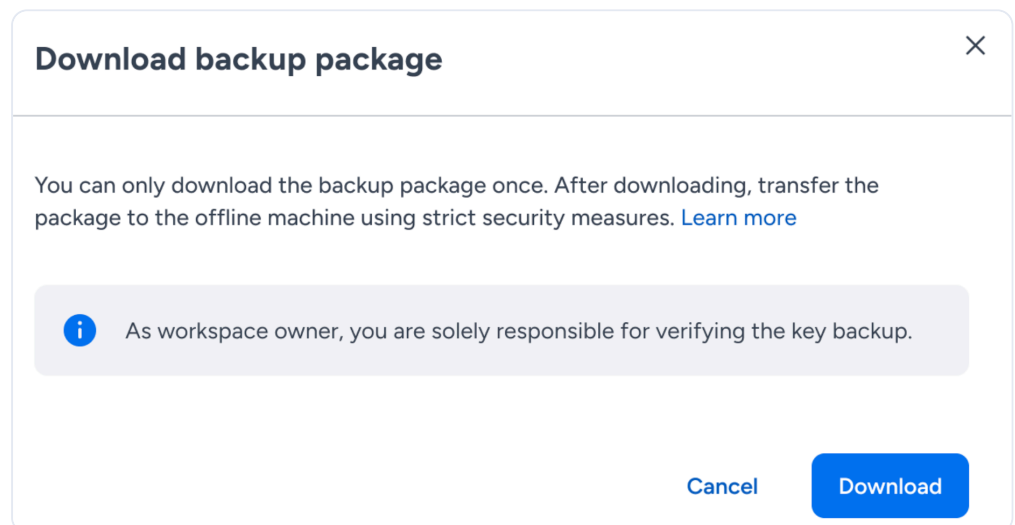
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vii. The Admin Quorum and Owner enter their Fireblocks mobile app PIN codes and complete biometric authorization to approve the request.

xiii. In your Console, the backup status updates to **Pending download**.



xiv. After finalizing the approval process on the Fireblocks mobile app, the workspace Owner can click **Download backup kit** to download the backup package. The backup package can only be downloaded once.



xv. After downloading, the Owner must transfer the package to an offline machine.

xvi. Once the backup kit has been transferred to the offline machine, the Owner can [verify the recovery package](#).

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Overview

Start the backup process

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